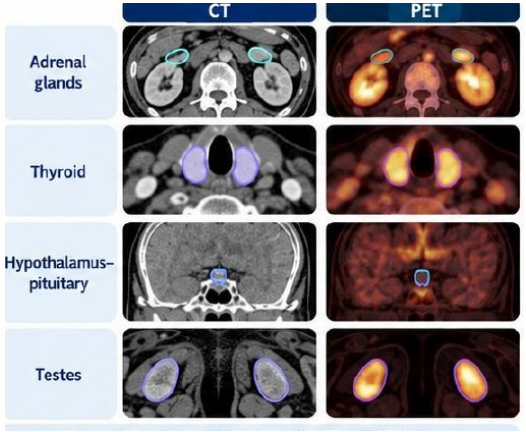
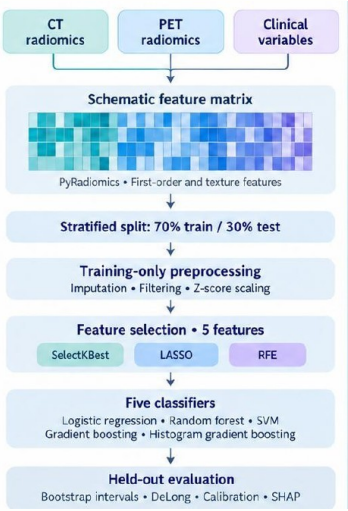
RADIOMICS · SYSTEMIC PHENOTYPING

Endocrine-organ PET/CT radiomics



Cohort: 101 men

- Recurrence, negative PSMA PET/CT
- 67 progressed, 34 progression-free
- Normal adrenal, thyroid, pituitary, testes



Fusion modelling

- CT + PET radiomics + clinical variables
- 70/30 split; 5 features; 5 classifiers
- Held-out test, DeLong, SHAP

Best model (test AUC)

0.758

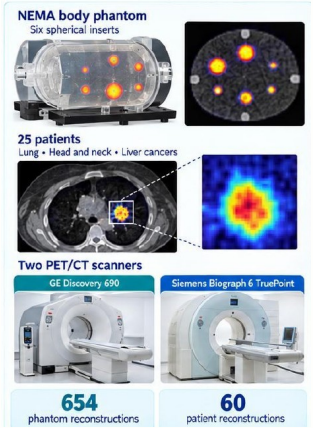
Testis CT + PET + clinical

Clinical only	**0.727**
CT + clinical	**0.729**
PET + clinical	**0.733**
CT + PET + clinical	**0.758**
ΔAUC vs clinical	**0.031**
DeLong p	**0.569**

Systemic signals beyond visible lesions add modest value — single centre, no external validation.

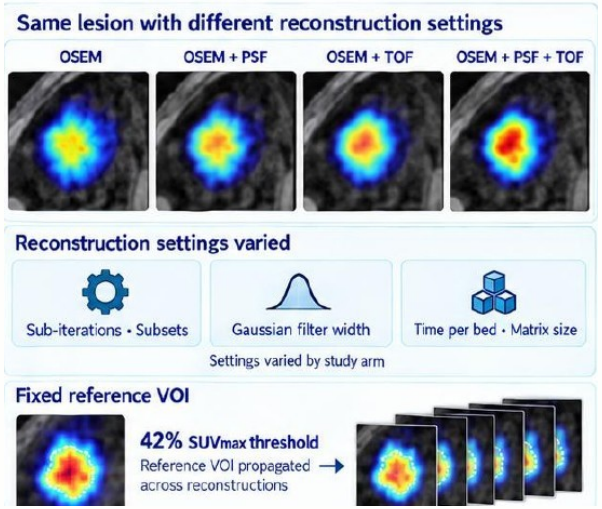
RADIOMICS · ROBUSTNESS

PET reconstruction changes radiomic features



Phantom + patients

- NEMA phantom and 25 patients
- 2 scanners: GE D690, Siemens Biograph
- 714 reconstructions in total



Same lesion, many settings

- OSEM ± PSF ± TOF, iterations, filter
- Time per bed and matrix size
- 100 features; variability as COV

Features robust (COV ≤5%)

47%

10% unstable (COV >20%)

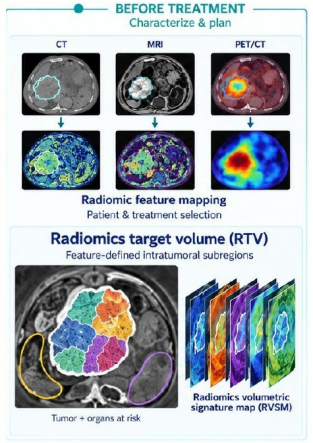
Geometry	**100%**
Intensity	**44%**
Texture	**41%**
Matrix size: COV >20%	**56%**
Matrix size: COV ≤5%	**6%**

Robustness is feature- and setting-dependent: standardize reconstruction and voxel size.

Shiri et al., Eur Radiol 2017 · doi:10.1007/s00330-017-4859-z

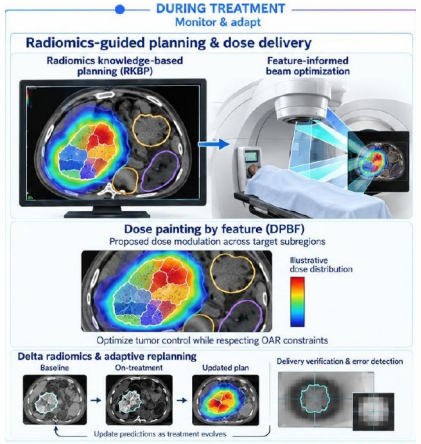
RADIOMICS-GUIDED RADIATION THERAPY

Radiomics across the radiotherapy timeline



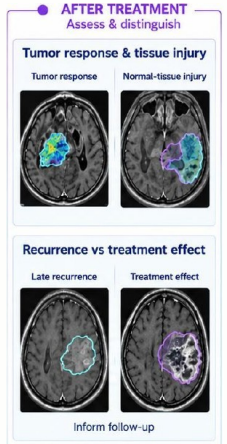
Before · characterize & plan

- Radiomic feature maps (CT, MRI, PET)
- Radiomics target volume (RTV)
- Signature map of tumour and organs



During · monitor & adapt

- Knowledge-based planning (RKBP)
- Dose painting by feature (DPBF)
- Delta radiomics for replanning



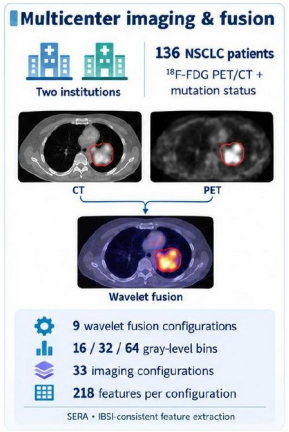
After · assess

- Response vs normal-tissue injury
- Recurrence vs treatment effect
- Guide follow-up

Inputs: longitudinal imaging + radiomics + clinical data + dose — needs prospective validation.

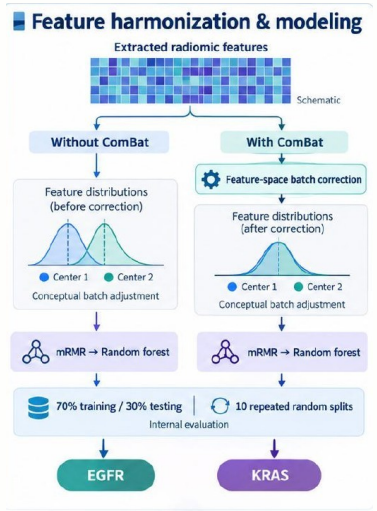
RADIOGENOMICS · MULTICENTRE HARMONIZATION

ComBat harmonization for EGFR and KRAS



136 NSCLC, two centres

- FDG PET/CT + mutation status
- Wavelet PET/CT fusion: 33 image sets
- 218 features per set



Harmonize and model

- With vs without ComBat
- mRMR → random forest
- 70/30 split × 10 repeats

Best fused model

AUC 0.94

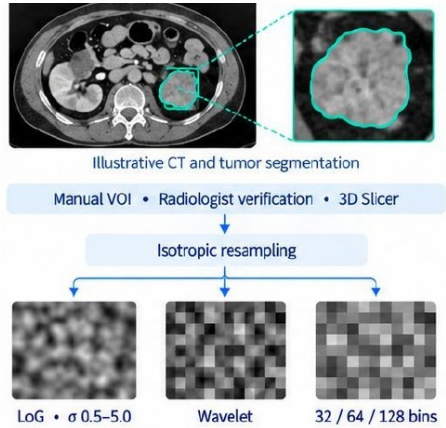
EGFR (KRAS 0.93)

EGFR before	**0.87–0.90**
EGFR after	**0.92–0.94**
KRAS before	**0.85–0.90**
KRAS after	**0.91–0.94**

Harmonization improved multivariate prediction ($q < 0.05$); single-feature effects varied.

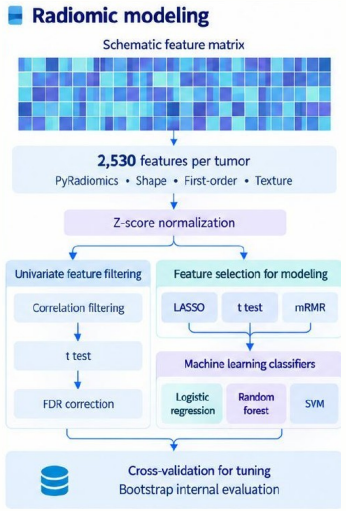
RADIOMICS · TUMOUR GRADING

CT radiomics for Fuhrman grade in ccRCC



71 ccRCC patients (TCIA)

- 31 low grade, 40 high grade
- Manual VOI, radiologist-verified
- LoG, wavelet and 32–128 bin images



Radiomic modelling

- 2,530 features per tumour
- LASSO / t-test / mRMR selection
- LR, random forest, SVM

Highest AUC

0.83

SVM + t-test + LoG σ 4.5

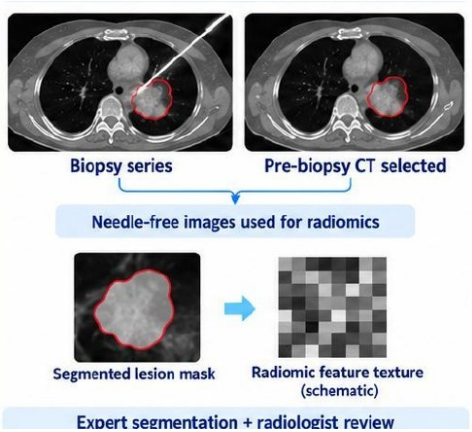
Original (SVM)	**0.76**
128 bins (SVM)	**0.77**
32 bins (LR)	**0.73**
Sig. texture features	**4**

Potential non-invasive preoperative grading — exploratory, no external validation.

Nazari et al., La Radiologia Medica 2020

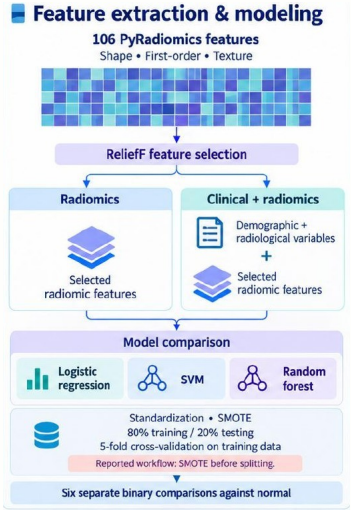
RADIOMICS · VIRTUAL BIOPSY

CT-based virtual biopsy of thoracic lesions



300 patients

- Pre-biopsy CT (needle-free)
- 189 tumoral, 111 normal
- 5 tumour types; pathology labels



Features and models

- 106 PyRadiomics features, ReliefF
- Radiomics vs clinical + radiomics
- LR, SVM, random forest

Best (vs normal)

AUC 0.92

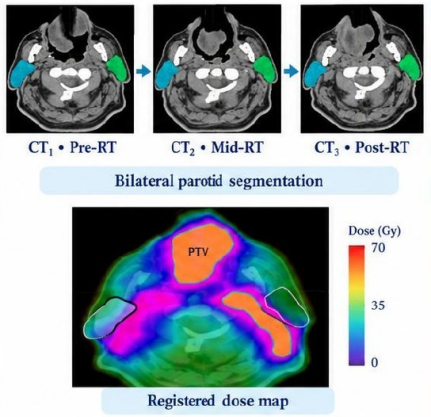
Lymphoma, combined model

Tumoral	0.72
Adenocarcinoma	0.74
SCC	0.82
SCLC	0.88
Mesenchymal	0.86

Potential decision support beside tissue biopsy — caution: SMOTE before splitting may inflate results.

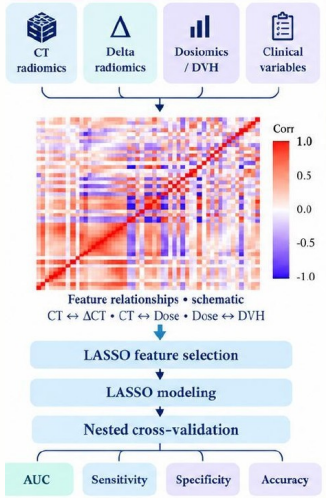
RADIOMICS · TOXICITY PREDICTION

Delta radiomics and dose for late xerostomia



31 patients, 59 parotids

- Head and neck VMAT, 58–70 Gy
- CT pre-, mid- and post-RT
- Registered dose map



Feature integration

- CT, delta, dosiomics/DVH, clinical
- LASSO feature selection
- Nested cross-validation

Highest AUC

0.89

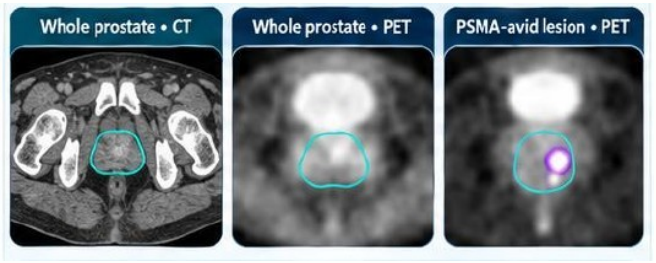
CT3 · ΔCT2−1 + dose · DVH

CT1 (pre-RT)	**0.68**
CT2 (mid-RT)	**0.84**
Dose only	**0.72**
Clinical	**0.88**

Longitudinal parotid imaging + dose supports xerostomia risk assessment.

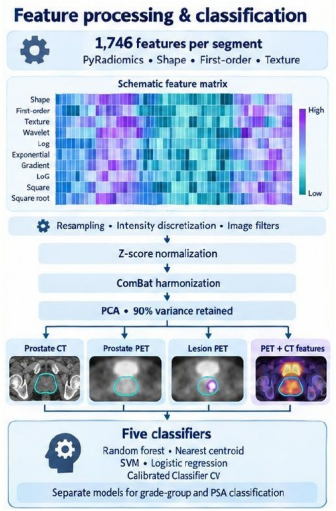
RADIOMICS · PSMA PET/CT

Multicentre PSMA PET/CT radiomics for risk



243 patients, 4 centres

- [68Ga]Ga-PSMA-11 PET/CT, untreated
- 186 development, 57 external test
- Whole prostate and PSMA-avid lesion



Processing and classifiers

- 1,746 features per segment
- ComBat + PCA (90% variance)
- 5 classifiers; GGG and PSA tasks

External accuracy

76.4%

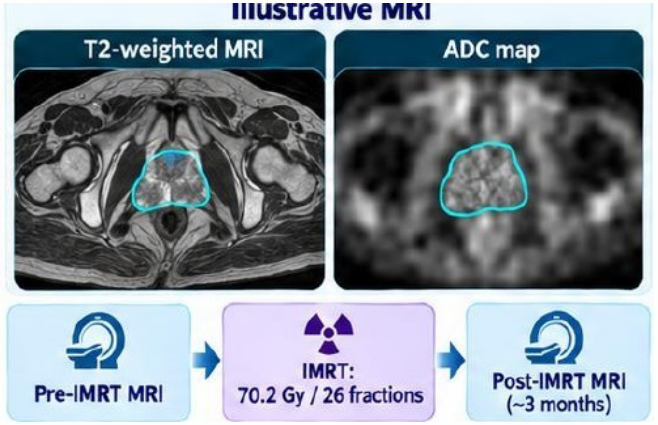
PSA <20 vs ≥20 (PET + CT)

GGG internal	**78.0%**
GGG external	**73.8%**
PSA internal	**75.4%**
PSA external	**76.4%**

PET lesion features and PET/CT fusion show potential for non-invasive risk stratification.

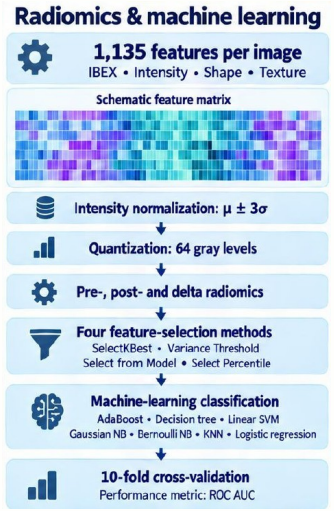
RADIOMICS · TREATMENT RESPONSE

MRI radiomics for IMRT response in prostate



33 patients, prospective

- T2-weighted MRI and ADC maps
- Before and ~3 months after IMRT
- Responder = ADC rise >20% (15 of 33)



Radiomics and ML

- 1,135 IBEX features
- Pre, post and delta radiomics
- 7 classifiers, 10-fold CV

Early response (AUC)

0.78

Best pre-treatment T2 model

Response, ADC	**0.77**
Gleason, T2	**0.74**
Gleason, ADC	**0.70**
Stage, ADC	**0.68**

Pre-treatment MRI radiomics carries response signals — exploratory, no external validation.

Abdollahi et al., La Radiologia Medica 2019

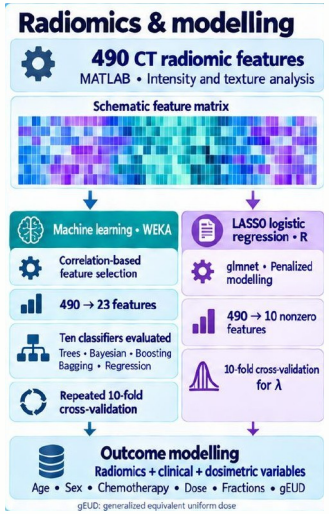
COCHLEAR CT RADIOMICS · TOXICITY

Predicting radiation-induced hearing loss



47 patients, 94 cochleae

- Head and neck 3D-CRT, 50–70 Gy
- Audiometry to 6 months
- 37 of 94 ears lost hearing (~40%)



Radiomics and modelling

- 490 CT features
- WEKA ML and LASSO regression
- Plus age, chemo, dose, gEUD

LASSO model (AUC)

0.885

10 selected features

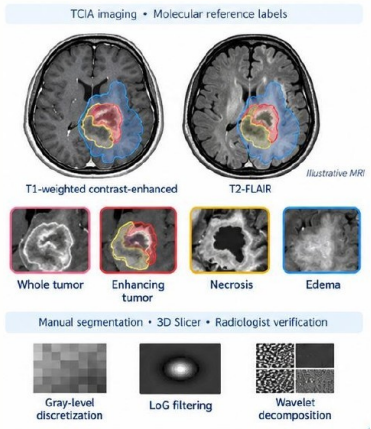
ML accuracy	**76.1%**
ML precision	**75.9%**
gEUD-only AUC	**0.79**

Adding gEUD improved precision and AUC — small cohort, needs validation.

Abdollahi et al., Physica Medica · doi:10.1016/j.ejmp.2017.10.008

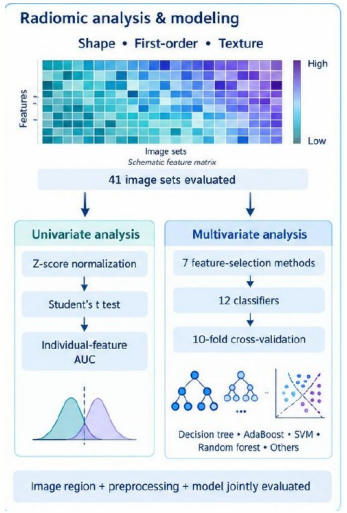
RADIOGENOMICS · GLIOBLASTOMA

MRI radiomics for MGMT promoter methylation



82 GBM patients (TCIA)

- 45 methylated, 37 unmethylated
- T1-CE and T2-FLAIR MRI
- Tumour, necrosis, edema regions



Radiomic analysis

- 41 image sets evaluated
- 7 selectors × 12 classifiers
- 10-fold cross-validation

Best model (AUC)

0.78

Decision tree · edema · LoG

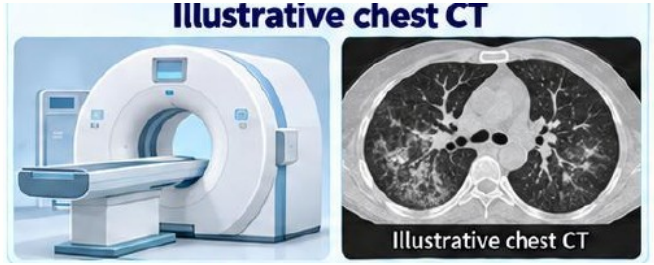
AdaBoost, edema	**0.74**
Best single feature	**0.71**

Strongest signal came from the edema region — exploratory, internal validation only.

Hajianfar et al., World Neurosurgery 2019

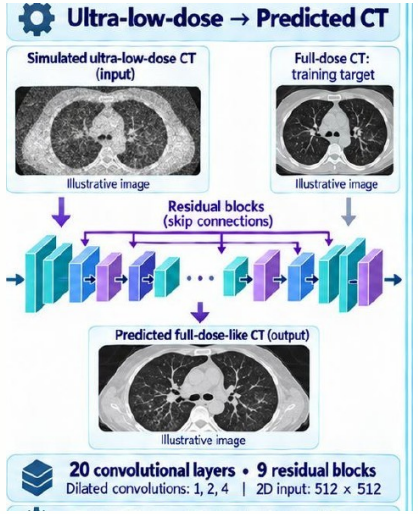
DEEP LEARNING · IMAGE QUALITY

Deep learning for ultra-low-dose chest CT



1,141 CT exams, 9 centres

- 312 COVID-19 positive
- Train 800 · test 170 · external 171
- Dose 6.5 → 0.72 mGy (−89%)



Deep residual network

- Ultra-low-dose → full-dose-like CT
- 20 conv layers, 9 residual blocks
- Quantitative + radiologist scoring

Radiologist score /5

4.42

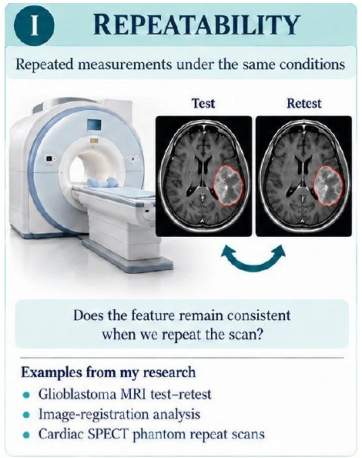
Up from 2.78 (full dose 4.72)

SSIM	**0.89 → 0.97**
PSNR (dB)	**29.4 → 33.6**
RMSE	**0.16 → 0.08**

Better image quality does not guarantee lesion fidelity — some outliers altered lesions.

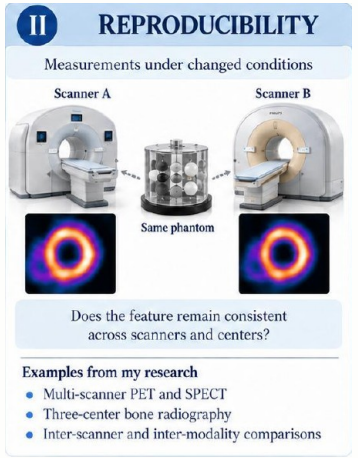
RADIOMICS · FEATURE RELIABILITY

Repeatability, reproducibility and robustness



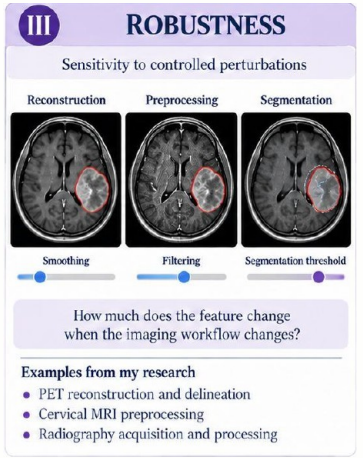
I Repeatability

- Same scanner, test–retest
- Glioblastoma MRI test–retest
- Cardiac SPECT phantom repeats



II Reproducibility

- Across scanners and centres
- Multi-scanner PET and SPECT
- Three-centre bone radiography



III Robustness

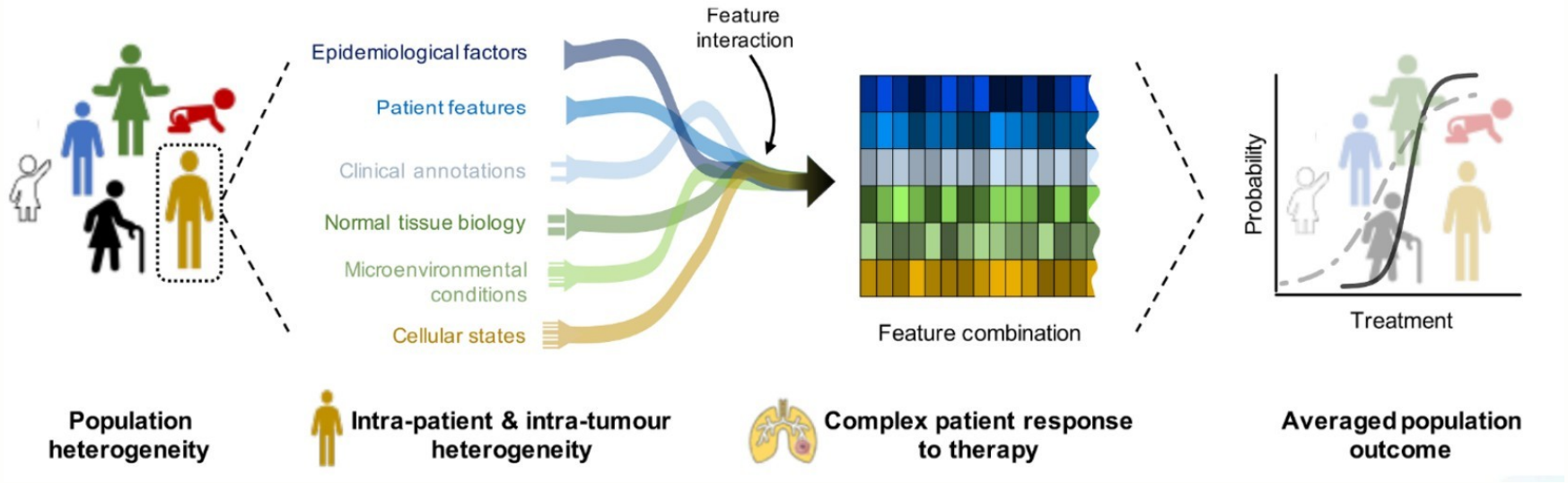
- Workflow perturbations
- PET reconstruction, delineation
- MRI preprocessing

Research aim: identify stable features and quantify the sources of variability.

Abdollahi et al., Phys Med Biol 2022

THE CHALLENGE

Treatment is designed for the average patient



Population heterogeneity
Every patient differs

Within-patient variation
Biology, microenvironment, cell states

Complex response
Features interact differently

Averaged outcome
The average curve fits almost no one

Averaging hides the individual patient → we need patient-specific models.

THE SOLUTION · DIGITAL TWINNING

Trials for patients who are not average



iScience

CellPress
OPEN ACCESS

Perspective

Designing clinical trials for patients who are not average

Thomas E. Yankeelov,[1,2,3,4,5,9,*] David A. Hormuth II,[4,5] Ernesto A.B.F. Lima,[4,6] Guillermo Lorenzo,[4,10] Chengyue Wu,[4] Lois C. Okereke,[4] Gaiane M. Rauch,[7] Aradhana M. Venkatesan,[7] and Caroline Chung[8]

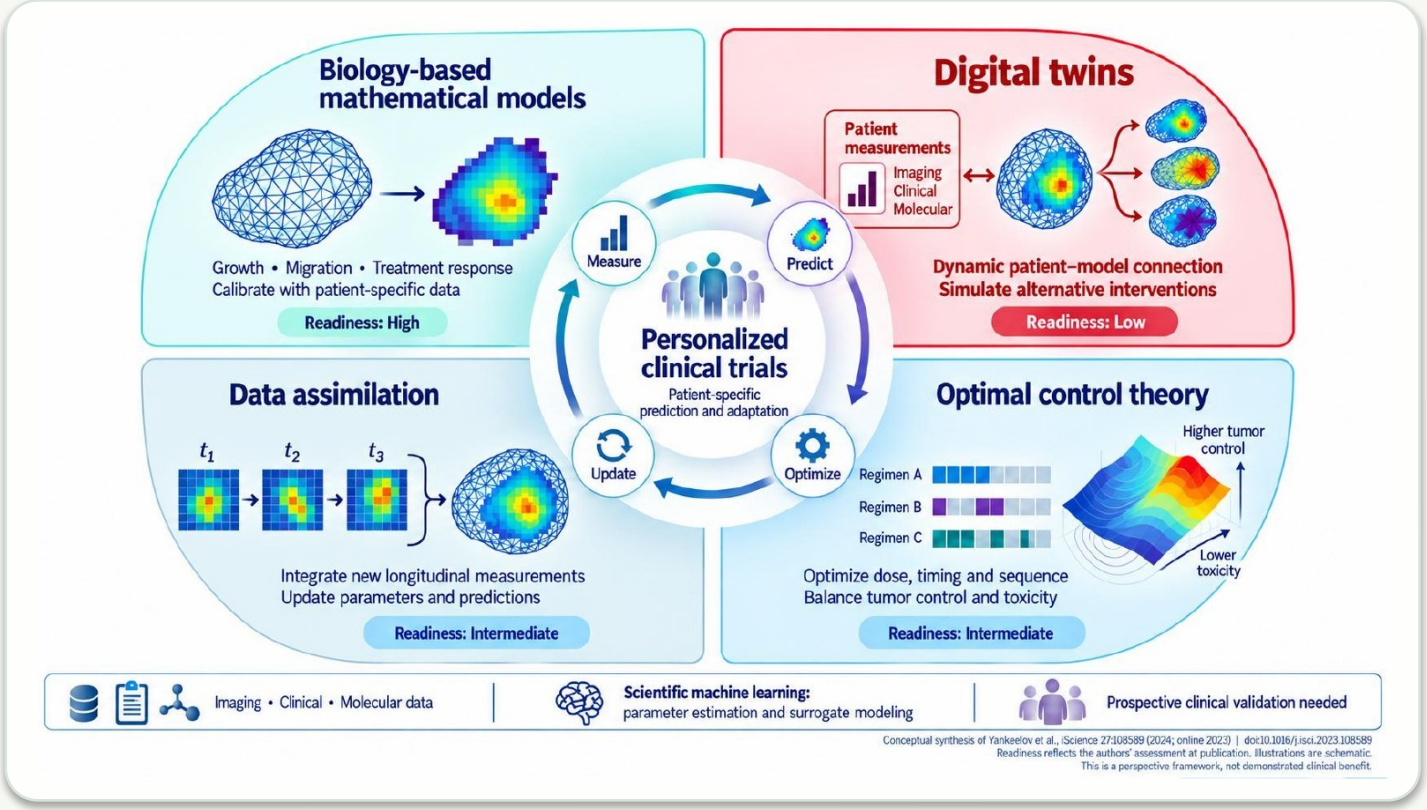
The idea

- Stop fitting each patient to a population curve
- Build a model of this patient
- Calibrate it with their own imaging and data
- Update it as treatment proceeds
- Use it to choose and adapt therapy

From averaged evidence to personalized, adaptive decisions.

THE SOLUTION · DIGITAL TWINNING

Four building blocks for personalized trials



Biology-based models
Growth and response, per patient
Readiness: high

Data assimilation
Update with new scans
Readiness: intermediate

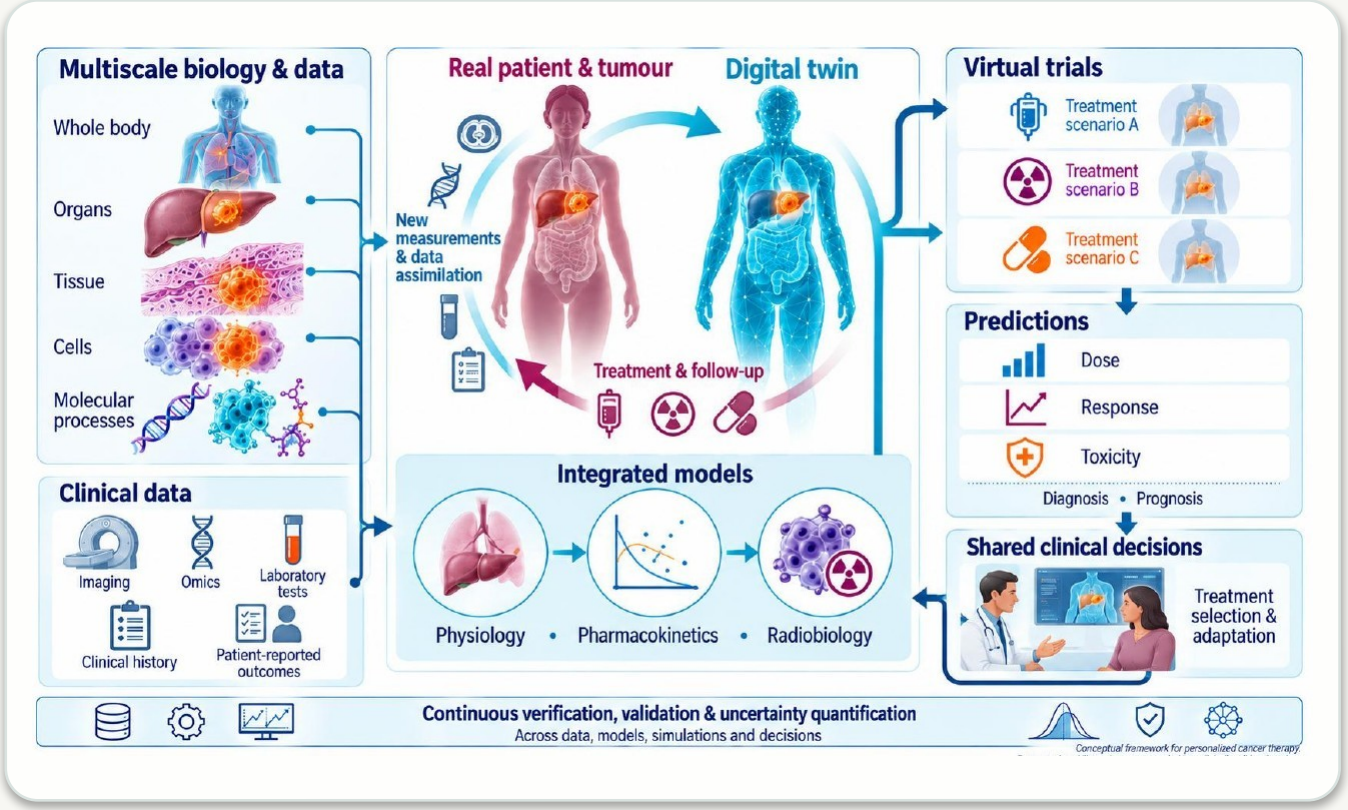
Optimal control
Best dose and timing vs toxicity
Readiness: intermediate

Digital twins
Live patient–model link
Readiness: low

Loop: measure → predict → optimize → update — needs prospective clinical validation.

THE SOLUTION · DIGITAL TWINNING

What is a digital twin?



Definition (NASEM 2024)

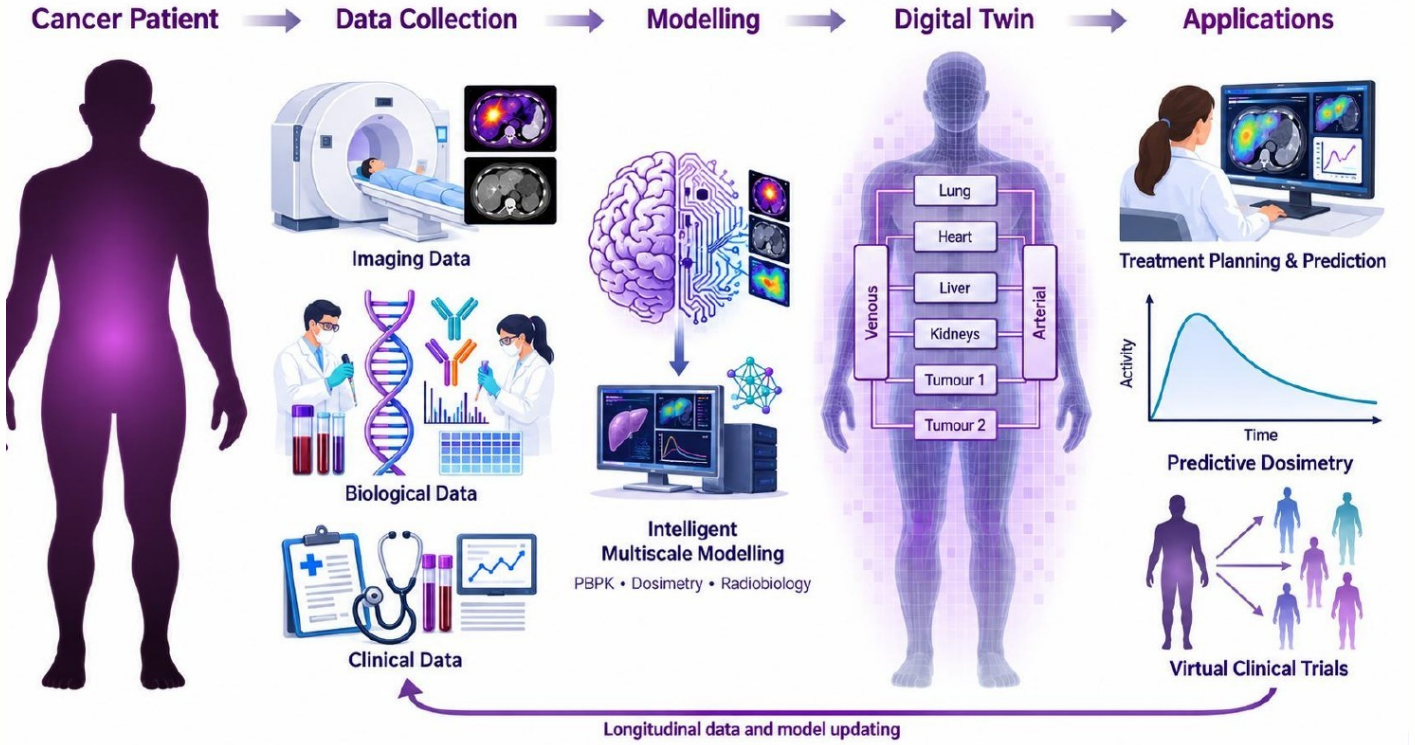
A virtual representation of a specific real-world system, dynamically updated with its data, that predicts its behaviour and informs decisions about it.

- Multiscale biology + clinical data
- Virtual trials of options A, B, C
- Predict dose, response, toxicity

Bidirectional link between patient and model, with continuous verification and uncertainty quantification.

THERANOSTIC DIGITAL TWINNING

From the patient to a theranostic digital twin



Data
Imaging, biological, clinical

Modelling
PBPK, dosimetry, radiobiology + AI

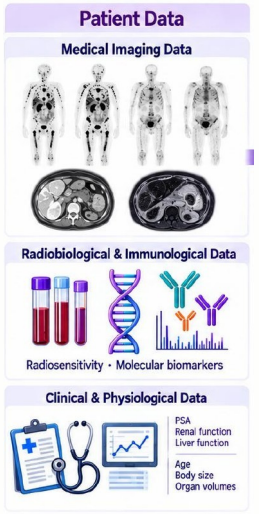
Digital twin
Whole-body organ and tumour compartments

Applications
Planning, predictive dosimetry, virtual trials

Longitudinal data keep updating the model.

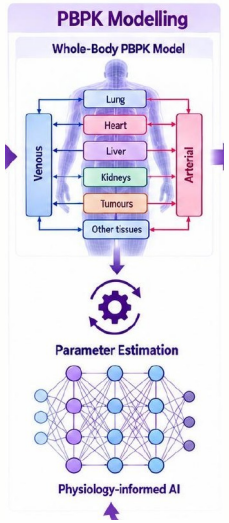
THERANOSTIC DIGITAL TWINNING

Building the twin: PBPK model for ^{177}Lu-PSMA



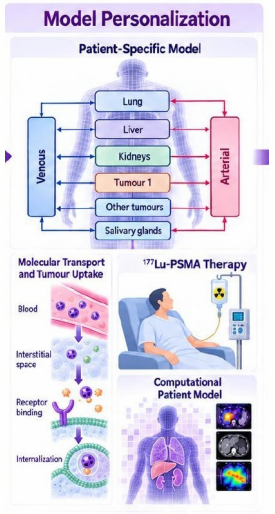
1 Patient data

- Imaging, radiobiology
- PSA, renal and liver function



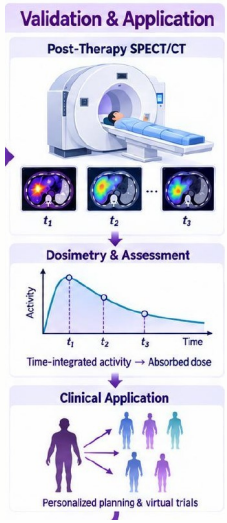
2 PBPK model

- Whole-body compartments
- Physiology-informed AI fits parameters



3 Personalization

- Patient-specific organs
- Binding and internalization



4 Validation & use

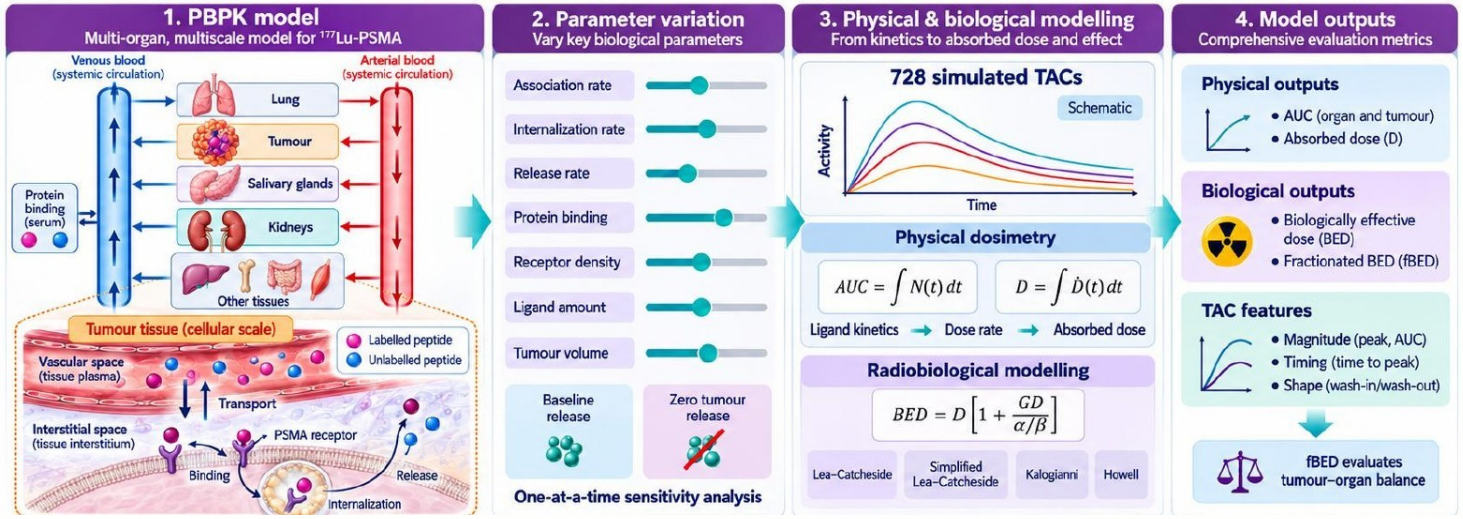
- Post-therapy SPECT/CT
- Absorbed dose; virtual trials

Model refinement with longitudinal data closes the loop.

Abdollahi et al., Theranostics 2024

THERANOSTIC DIGITAL TWINNING

Model-informed ^{177}Lu-PSMA optimization



Association rate ↑
Tumour dose ↑, organs too

Internalization
Non-linear; peak at mid rates

Tumour release ↑
Tumour dose and BED ↓

Protein binding ↑
Dose ↓ in all tissues

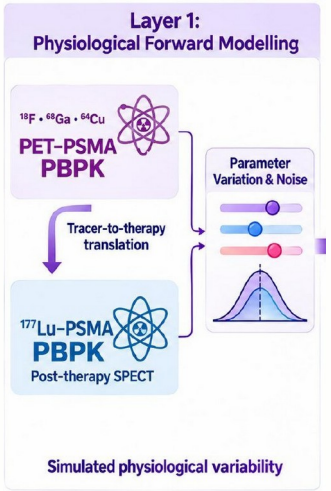
Receptor density ↑
Tumour dose ↑, organs ~same

Tumour volume ↑
Uptake ↑, tumour dose ↓

Patient-specific kinetics guide the tumour–organ trade-off (728 simulated TACs).

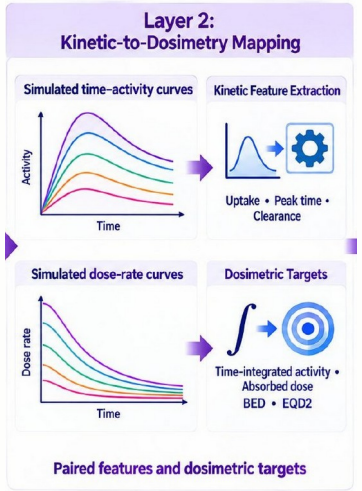
APPLICATION · PREDICTIVE DOSIMETRY

Predicting ^{177}Lu dosimetry from PET kinetics



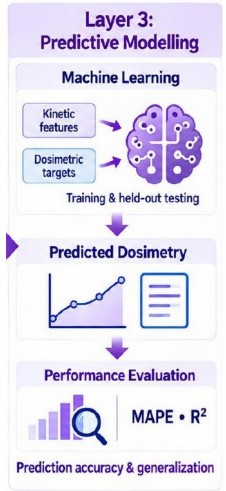
Layer 1 · Forward modelling

- PET-PSMA PBPK (^{18}F, ^{68}Ga, ^{64}Cu)
- Translated to ^{177}Lu-PSMA
- Parameter variation + noise



Layer 2 · Kinetics → dose

- Kinetic features: uptake, peak, clearance
- Targets: TIA, dose, BED, EQD2



Layer 3 · Prediction

- ML with held-out testing
- Evaluated by MAPE and R^2

Result: AMax (peak uptake) leads prediction; noise matters most in small tumours.